

- Y = output, income
- K = capital
- L = labor
- w = wage
- r = interest rate
- s = saving rate
- n = population growth rate
- d = depreciation rate
- N = population
- A = number of ideas
- Lowercase = per worker or person
 - Example: $y = Y/L$ = output per person
- $g(\text{---})$ = growth rate
- $\bar{z}$ = efficiency/productivity of idea creation
 - Example: constant theft of ideas decreases $\bar{z}$
- ℓ = share of scientists (ex: 0.4 is 40% of population are scientists)
- GDP (Gross Domestic Product):
 - Domestically produced goods THIS year
 - $Y = C + I + G + NX$
 - Nominal: Output in 2026 (or present day) dollars.
 - Real: Output in constant price level.
 - Real = Nominal GDP / Price Index
- Profit maximization
 - $MPK = r$
 - $MPL = w$
- Growth rates
 - $y_t = y_0(1 + g)^t$
 - $L_t = L_0(1 + n)^t$
 - $g = (y_t / y_0)^{1/t} - 1$
- Doubling time for $y = 70 / g$
- Log rules
 - If $z = x / y$, then $g(z) = g(x) - g(y)$
 - If $z = xy$, then $g(z) = g(x) + g(y)$
 - If $z = x^a$, then $g(z) = a(g(x))$
- Cobb-Douglas
 - $Y = A(K)^{1/3} (L)^{2/3}$
 - $y = Ak^{1/3}$
 - Constant returns to scale (sum of exponents = 1)
 - Diminishing marginal productivity of labor and capital (each exponent < 1)

- Know how to draw the graph
- Solow Model:
 - Parameters: s, n, d, A
 - Steady state k^* when $sy = (n+d)k$
 - Draw the graph with k on x-axis and y on y-axis
 - Understand where to put wage on the graph
 - Know how the graph changes with a shift in:
 - L
 - K
 - A
 - n
 - s
 - d
 - Steady state k^* and y^* only changes when parameters do
 - BUT in the overall (not per person) graph - a change in L can shift K^*
 - Transition state dynamics:
 - Growth rate is based on how far your k is from your k^* ! The further away, the faster growth
- Golden Rule:
 - Maximize consumption when $MPK = n + d$
- Romer Model: importance of ideas (A)
 - $g[A] = \bar{z}\ell N$
 - $A_t = A_0(1+g[A])^t$
 - $y_t = A_0(1 - \ell)(1+g[A])^t$
 - Draw the graph (ratio scale)
 - Slope = $g[A]$
 - Know how it changes with a shift in:
 - The number of workers
 - The number of scientists
 - The share of workers
 - The share of scientists
- Big lessons from the class
 - Ideas are more important than capital accumulation for growth
 - Institutions are critical for growth - encourage idea creation and investment